

Causal Offset Bounded TE Curve Verification Screen Closeout

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This closeout records the bounded TE Curve Verification Pipeline screen for the causal_offset_mean_calibration_pilot_2026_07_22 forward candidates on polished_dataset setpoint inputs. The screen compared the two causal offset / mean calibration candidates against the accepted polished-setpoint forward baselines and the recent scalar shape-objective high-water reference.

Execution Summary

Item	Value
Screen	causal_offset_bounded_track2_screen_2026_07_22
Dataset	polished_dataset
Input mode	setpoints
Surface	Fw / forward
Candidate count	5
Curve count	100
Matrix output	output/validation_checks/track2_reference_comparison/2026-07-23-13-07-44_causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_causal_offset_bounded_track2_screen_polished_setpoints_fw
Matrix report	doc/reports/analysis/validation_checks/te_curve_verification_pipeline/2026-07-23-13-08-58_causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_causal_offset_bounded_track2_screen_polished_setpoints_fw_report.md
Reranker output	output/validation_checks/shape_gated_te_curve_reranker/2026-07-23-13-09-03_shape_gated_te_curve_reranker
Reranker report	doc/reports/analysis/te_curve_verification_pipeline/03_cvp_diagnostics/shape_gated_reranker/[2026-07-22]/causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_shape_gated_te_curve_reranker_report.md
Plot count	0

Metric Ranking

Rank	Candidate	Raw MAE [deg]	RMSE [deg]	Mean Error [%]	P95 Error [%]
1	polished_setpoints_periodic_gru_sequence_Fw	0.001837	0.002179	3.750629	8.248098
2	polished_setpoints_periodic_mlp_harmonic_Fw	0.001938	0.002276	3.987765	10.017324
3	shape_objective_periodic_mlp_harmonic_Fw	0.002035	0.002412	4.222146	10.956483
4	causal_offset_mean_periodic_mlp_harmonic_Fw	0.002075	0.002478	4.314268	9.717112
5	causal_offset_mean_gru_sequence_Fw	0.002392	0.002868	5.009841	10.269009

Shape-Gated Decision

Rank	Candidate	Centered MAE [deg]	Shape Pass	Composite	Decision
1	polished_setpoints_periodic_gru_sequence_Fw	0.001483	0.950000	0.007008	recommended
2	polished_setpoints_periodic_mlp_harmonic_Fw	0.001490	0.920000	0.239545	candidate
3	shape_objective_periodic_mlp_harmonic_Fw	0.001578	0.910000	0.419662	candidate
4	causal_offset_mean_periodic_mlp_harmonic_Fw	0.001652	0.920000	0.424363	candidate
5	causal_offset_mean_gru_sequence_Fw	0.002024	0.000000	0.961925	shape gate failed

Harmonic Breakdown

Rank	Candidate	FFT Sim.	Amp Err. [%]	Phase Err. [deg]
1	polished_setpoints_periodic_gru_sequence_Fw	0.984971	17.554759	12.349713
2	polished_setpoints_periodic_mlp_harmonic_Fw	0.983739	15.623668	12.951608
3	shape_objective_periodic_mlp_harmonic_Fw	0.983709	18.390567	17.439607
4	causal_offset_mean_periodic_mlp_harmonic_Fw	0.983672	19.654545	18.516624
5	causal_offset_mean_gru_sequence_Fw	0.975176	77.578652	71.215158

Interpretation

The bounded screen does not promote either causal offset / mean calibration candidate.

The accepted `polished_setpoints_periodic_gru_sequence_Fw` baseline remains the forward recommendation. It leads on raw MAE, centered MAE, shape pass rate, and composite score.

The non-windowed causal offset MLP remains a valid completed pilot, but it ranks behind the accepted GRU, the non-windowed polished-setpoint MLP baseline, and the prior shape-objective MLP reference. It does not justify expansion from this profile.

The time-windowed causal residual-offset GRU is rejected for this branch. Its shape pass rate is `0.000000`, and the reranker reports failures dominated by harmonic amplitude error, phase error, derivative agreement, FFT similarity, and peak-to-peak error.

Output Quality Finding

The matrix summary reports `report_plot_count: 0` with empty preview and report plot lists. Therefore this screen has scalar and shape-gated evidence, but it does not have the expected Track 2 measured-versus-predicted plot package.

The local operator log `.temp/campaign_log.log` also contains very long unwrapped transfer/progress lines and PowerShell host records. The most visible source is interactive `scp` transfer progress emitted during remote sync. This is an operator-output problem, not a model-result change, and should be fixed permanently in the campaign launcher/tooling after this closeout commit.

Decision

Do not promote the causal offset / mean calibration candidates.

Do not expand the causal residual-offset GRU branch from this evidence.

Keep both time-windowed and non-windowed roads as comparison categories for future designs, but close this direct causal-offset mean-calibration profile as non-promoted.

The next repository step is to repair bounded campaign output ergonomics: remote sync logs should not flood the terminal with unwrapped progress lines, and bounded Track 2 screens should reliably produce the expected measured-versus predicted TE curve plots when plot generation is part of the operator-facing reporting contract.